

Denovo comparison: Instanovo vs Casanovo

26 May, 2025

Sample overview

Sample	Replicate	Comment	Sample complexity
1	1	1 Ab in the lowest concentration (1ng) + blood background	hard
2	1	1 Ab in the lowest concentration (1ng) + blood background	hard
3	1	1 Ab in the lowest concentration (1ng) + blood background	hard
4	1	1 Ab in the lowest concentration (1ng) + blood background	hard
5	1	1 Ab in the lowest concentration (1ng) + blood background	hard
6	1	1 Ab in the lowest concentration (1ng) + blood background	hard
39	1	1 Ab in the highest concentration (1000ng), no blood background	easy
43	1	1 Ab in the highest concentration (1000ng), no blood background	easy
47	1	1 Ab in the highest concentration (1000ng), no blood background	easy
70	1	all 6 Abs in the highest concentration (1000ng) + blood background	hard

Comparison of coverage Instanovo vs Casanovo

Heavy chain

mAb name	Sample	Vregion_Casanovo	Vregion_Instanovo	Vregion_Instanovo_v1.1
h9C12-Q97A_HC	1	0.00	6.84	6.84
h9C12-WT_HC	2	0.00	0.00	0.00
Brimab_HC	3	28.70	31.30	35.65
Umab_HC	4	13.45	13.45	13.45
PGT121_HC	5	8.33	8.33	0.00
PGDM1400_HC	6	0.00	0.00	0.00
h9C12-WT_HC	39	52.99	49.57	84.62
Umab_HC	43	21.85	31.93	21.85
PGDM1400_HC	47	16.31	0.00	7.80
PGDM1400_HC	70	7.09	0.00	0.00
PGT121_HC	70	8.33	8.33	8.33
Umab_HC	70	0.00	0.00	0.00
Brimab_HC	70	0.00	0.00	0.00
h9C12-WT_HC	70	0.00	0.00	0.00
h9C12-Q97A_HC	70	0.00	0.00	0.00

mAb name	Sample	CDR3_Casanovo	CDR3_Instanovo	CDR3_Instanovo_v1.1
h9C12-Q97A_HC	1	0.00	0.00	0.00
h9C12-WT_HC	2	0.00	0.00	0.00

Brimab_HC	3	0.00	0.00	0.00
Umab_HC	4	0.00	0.00	0.00
PGT121_HC	5	0.00	0.00	0.00
PGDM1400_HC	6	0.00	0.00	0.00
h9C12-WT_HC	39	76.92	46.15	100.00
Umab_HC	43	21.43	21.43	21.43
PGDM1400_HC	47	2.78	0.00	2.78
PGDM1400_HC	70	0.00	0.00	0.00
PGT121_HC	70	0.00	0.00	0.00
Umab_HC	70	0.00	0.00	0.00
Brimab_HC	70	0.00	0.00	0.00
h9C12-WT_HC	70	0.00	0.00	0.00
h9C12-Q97A_HC	70	0.00	0.00	0.00

Light chain

mAb name	Sample	Vregion_Casanovo	Vregion_Instanovo	Vregion_Instanovo_v1.1
h9C12-LC	1	0.00	0.00	0.00
h9C12-LC	2	0.00	0.00	0.00
Brimab_LC	3	0.00	11.71	11.71
Umab_LC	4	7.48	7.48	24.30
PGT121_LC	5	11.43	11.43	11.43
PGDM1400_LC	6	0.00	0.00	0.00
h9C12_LC	39	19.63	19.63	42.99
Umab_LC	43	0.00	0.00	48.60
PGDM1400_LC	47	18.75	21.43	38.39
PGDM1400_LC	70	0.00	0.00	0.00
PGT121_LC	70	7.62	7.62	19.05
Umab_LC	70	0.00	0.00	37.38
Brimab_LC	70	0.00	0.00	0.00
h9C12_LC	70	0.00	0.00	0.00
h9C12_LC	70	0.00	0.00	0.00

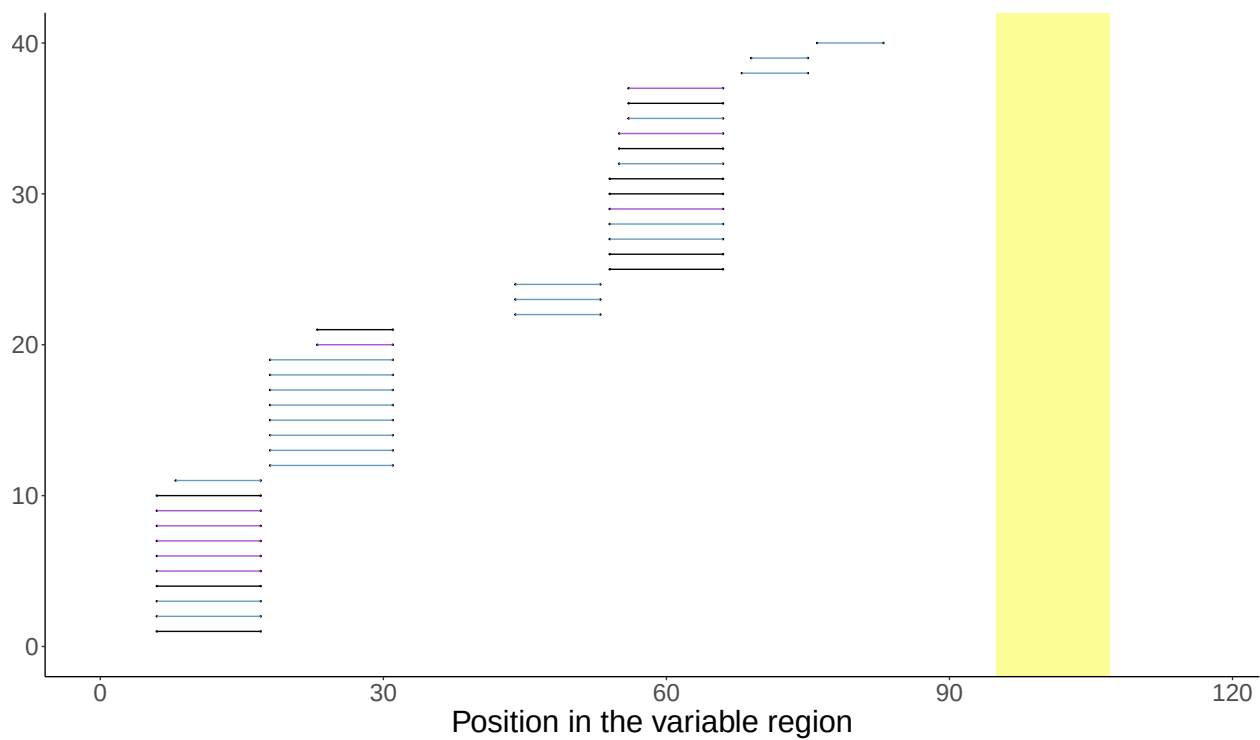
mAb name	Sample	CDR3_Casanovo	CDR3_Instanovo	CDR3_Instanovo_v1.1
h9C12-LC	1	0.00	0.00	0.00
h9C12-LC	2	0.00	0.00	0.00
Brimab_LC	3	0.00	0.00	0.00
Umab_LC	4	0.00	0.00	0.00
PGT121_LC	5	21.43	21.43	21.43
PGDM1400_LC	6	0.00	0.00	0.00
h9C12_LC	39	0.00	0.00	0.00
Umab_LC	43	0.00	0.00	0.00
PGDM1400_LC	47	54.55	54.55	100.00
PGDM1400_LC	70	0.00	0.00	0.00
PGT121_LC	70	0.00	0.00	0.00
Umab_LC	70	0.00	0.00	0.00
Brimab_LC	70	0.00	0.00	0.00
h9C12_LC	70	0.00	0.00	0.00

h9C12_LC	70	0.00	0.00	0.00
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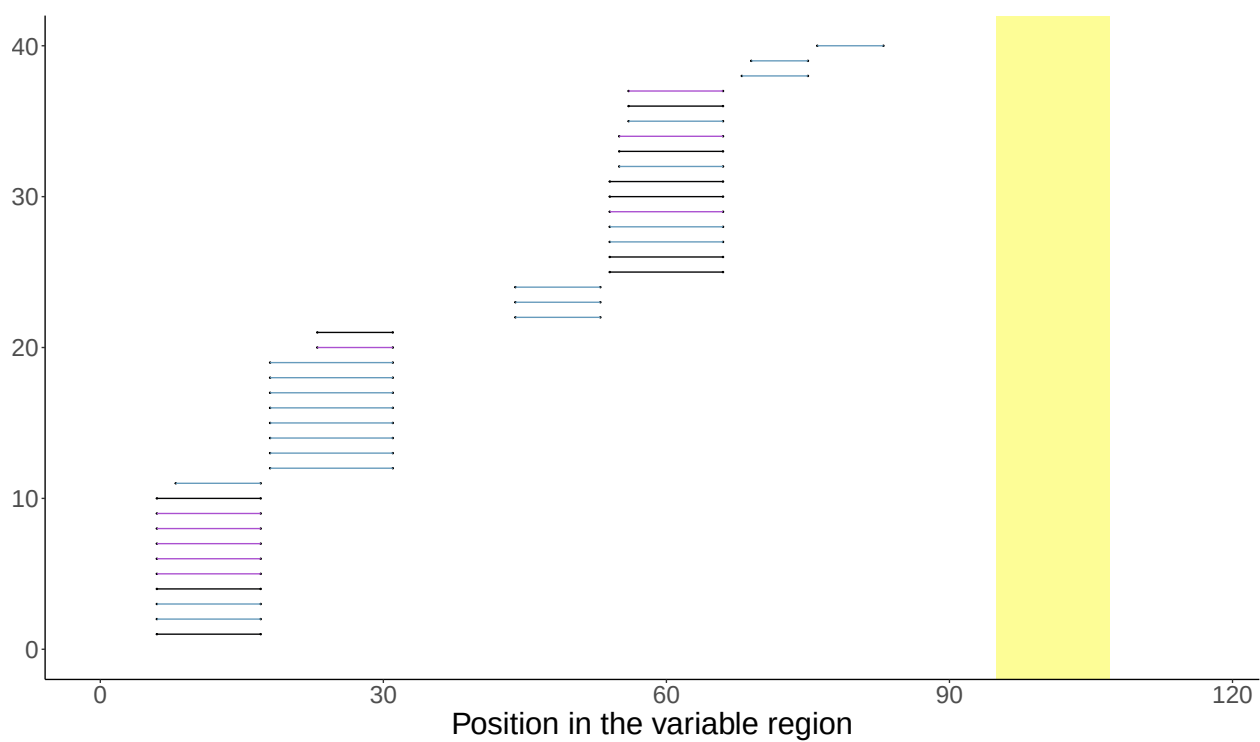
Based on coverage percentage, Instanovo v1.1 resulted in improved sequence coverage compared to the previous version. The previous version of Instanovo performed slightly worse than Casanovo, but Instanovo v1.1 performed slightly better.

Heavy chain V region coverage by denovo tool: Casanovo (violet),
Instanovo (black), Instanovo v1.1 (blue)

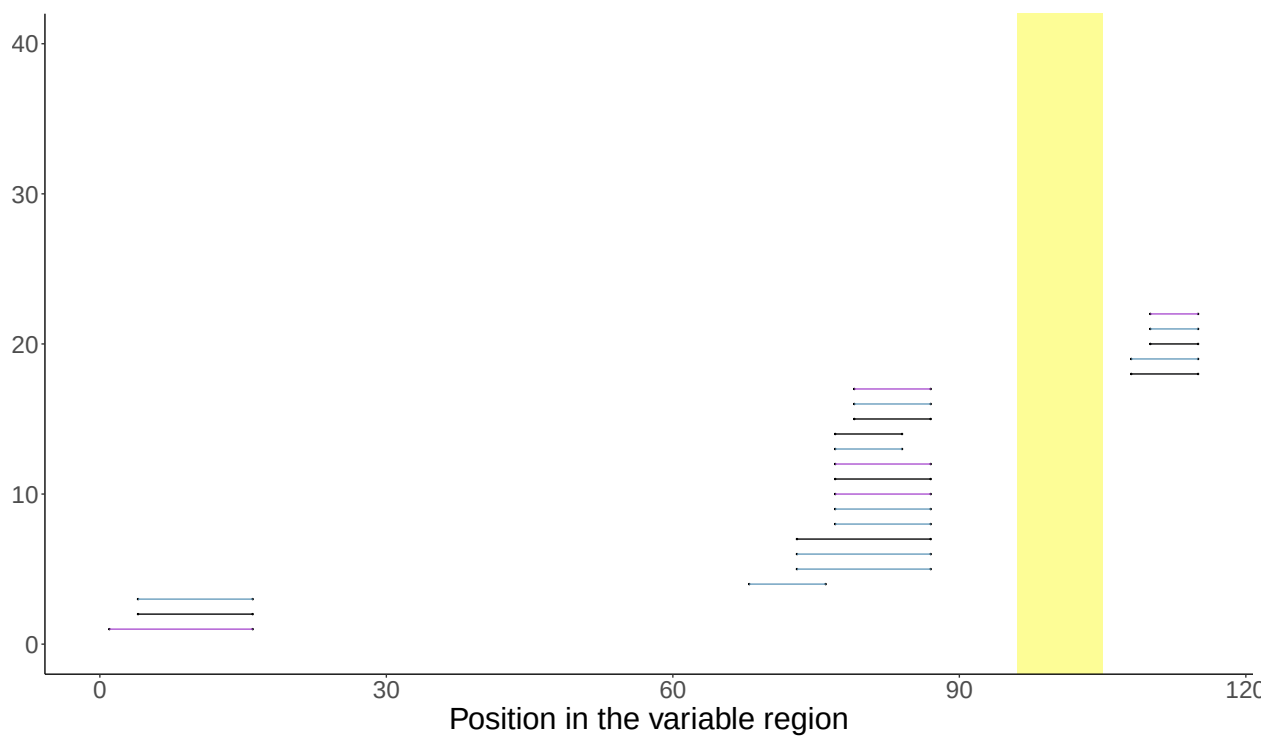
h9C12-WT_HC



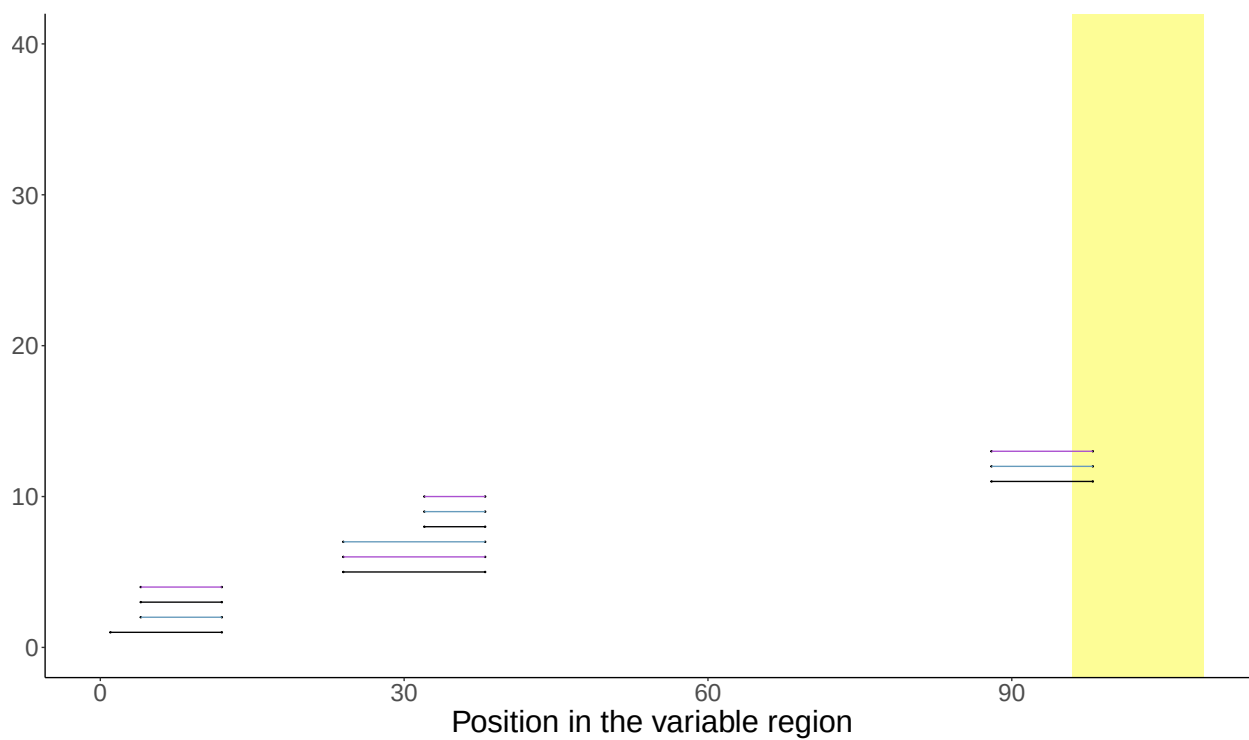
h9C12-Q97A_HC



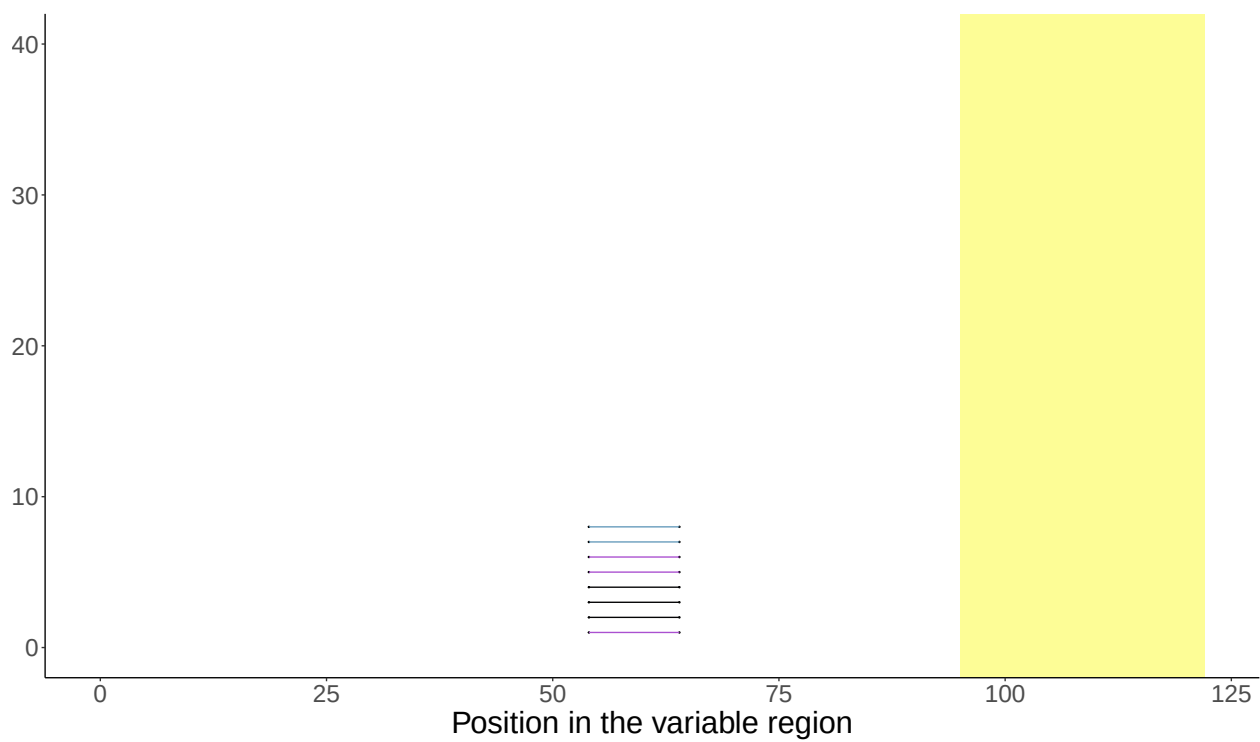
Brimab_HC



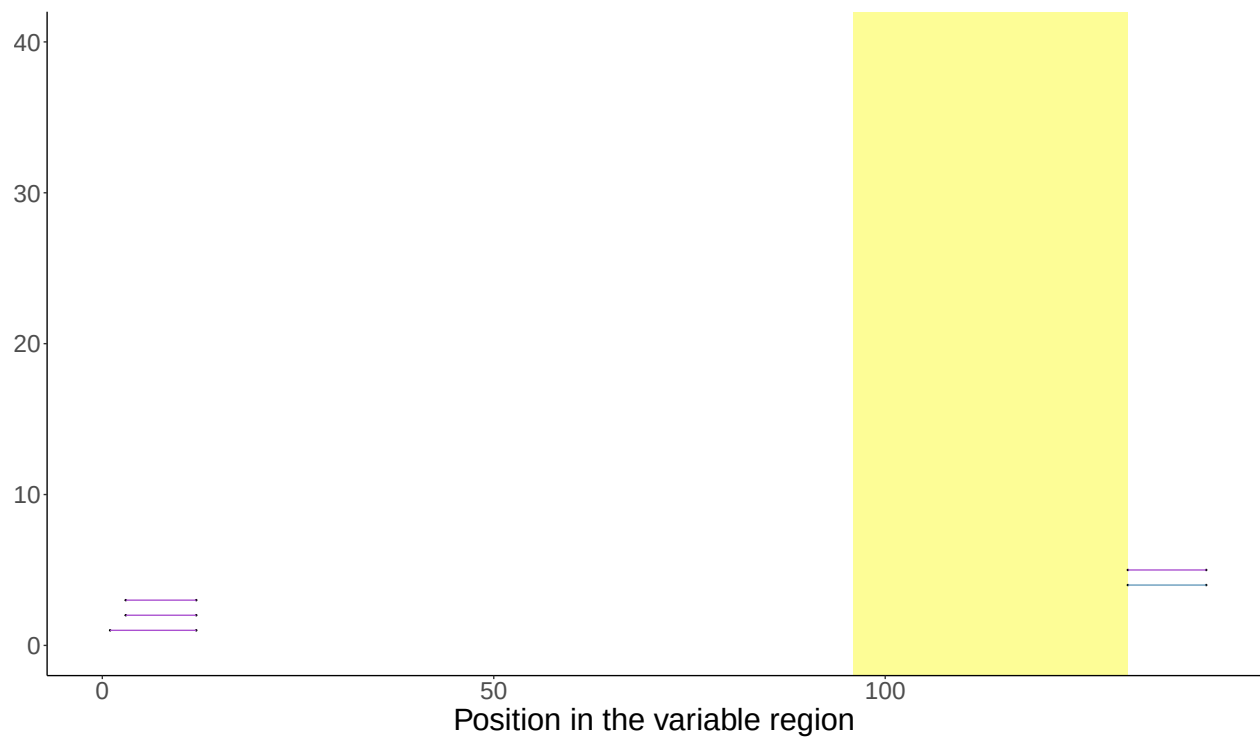
Urab_HC



PGT121_HC



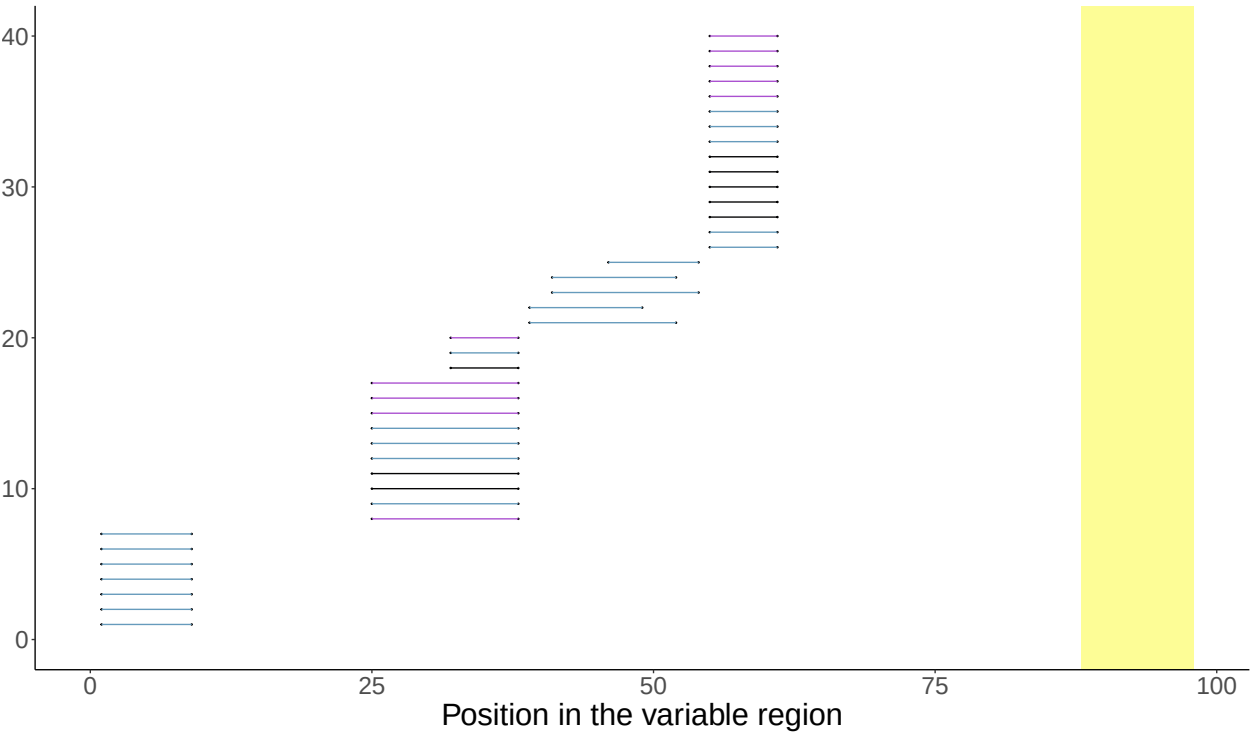
PGDM1400_HC



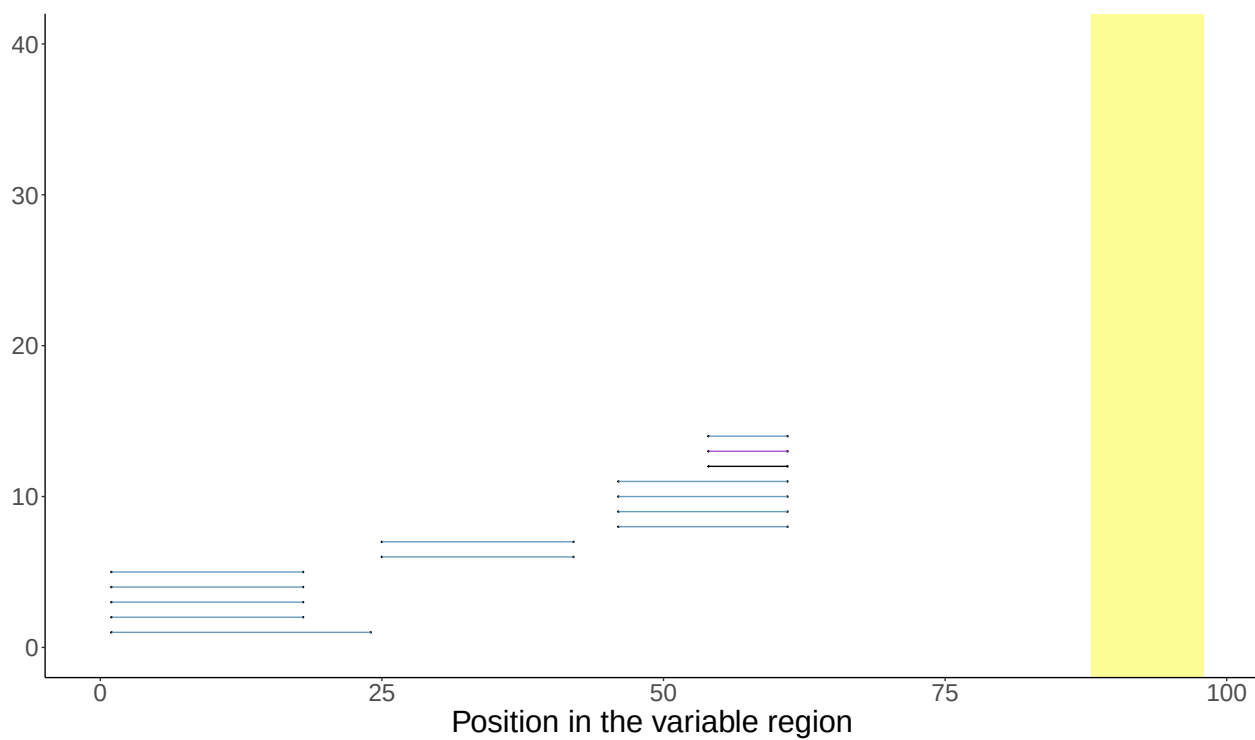
Overall, Casanovo (violet) and Instanovo (black) inferred mostly identical peptides, with Casanovo inferred some peptides not seen in Instanovo, and a few exceptions where Instanovo peptides are slightly shorter/longer than their Casanovo counterparts. Instanovo v1.1 (blue) performed much better than the previous version, and slightly outperform Casanovo in some cases.

Light chain V region coverage by denovo tool

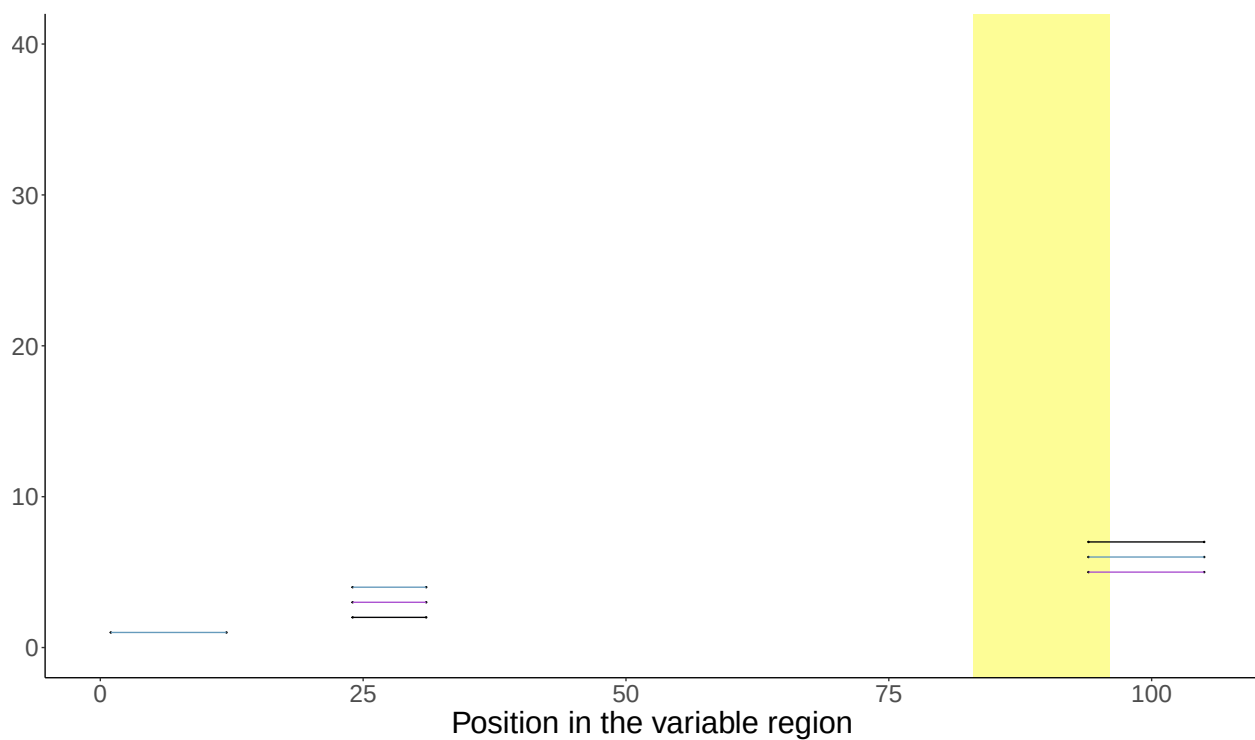
h9C12_LC



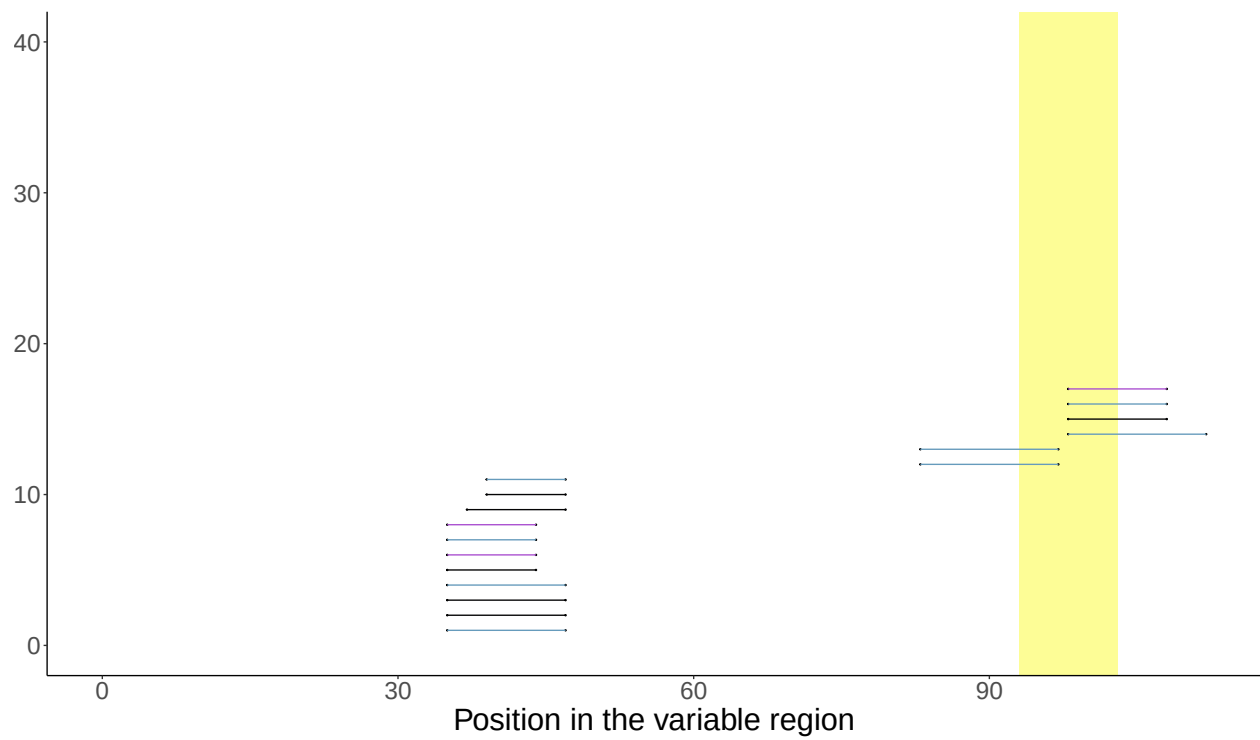
Umab_LC



PGT121_LC



PGDM1400_LC



Similar patterns can be observed on the light chain, with the exception of Umab where only Instanovo v1.1 performed noticeably better.